

Faculty of Information Technology

Department of Computer Science

Course: AI Programming

Lab Number: Lab 1

Lab Title: Introduction to Python Expressions and Built-in Functions

1. Objective

The purpose of this lab is to introduce students to the basics of Python programming and provide hands-on practice with Python arithmetic expressions, built-in functions and type conversion, mathematical functions from the math module, random number generation using the random module, and different uses of the print() function.

2. Required Software

Install Python 3 on your computer from: <https://www.python.org/downloads/>

During installation ensure that 'Add Python to PATH' is selected.

You may use Python IDLE, Visual Studio Code, PyCharm, or any Python-compatible editor.

3. Lab Instructions

Create ONE Python program that demonstrates the tasks listed in this lab. The file name must follow the format:

lab1_St#.py

Example: lab1_320240605022.py

4. Part 1 – Arithmetic Expressions

Evaluate and print the results of the following expressions:

`2 * 3`

`2 ** 3`

`5 + 2 * 5`

`(5 + 2) * 5`

`-4 - -4 - -4`

`2 ** 2 ** 0`

`(2 ** 2) ** 0`

`6 // 2`

`6 // 4`

`6.0 / 4.0`

`2.0 // 2.5`

`9.0 * 0.5`

`9.0 ** 0.5`

`6 % 2`

```
8 % 3
6.2 % 4
```

5. Part 2 – Type Conversion and Built-in Functions

Evaluate and print the results of the following:

```
float(4)
int(5.3)
float(int(5.3))
int(5.7)
float(7) // 4
int(7 / 4)
type(4.5)
type(7)
str(25)
```

6. Part 3 – Using the math Module

Import the math module and evaluate the following:

```
math.floor(7.8)
math.ceil(7.2)
round(6.58)
round(6.583,2)
abs(-15)
pow(2,4)
math.sqrt(81)
```

7. Part 4 – Random Number Generation

Import the random module and generate values using:

```
random.randint(1,10)
random.randint(50,100)
random.random()
random.randrange(0,20,2)
```

8. Part 5 – Using the print() Function

Demonstrate different ways to use the print() function including:

- Normal printing
- Printing multiple values
- Using sep parameter
- Using end parameter
- Printing text and numbers together.

9. Part 6 – Formatted Output

Create a formatted section displaying student information such as name, student ID, and major.

10. Deliverables

Submit ONE Python file containing all parts of the lab.

File name format: lab1_St#.py

Example: lab1_320240605022.py

11. Submission Instructions

Submit the Python file through Moodle before the specified deadline.

12. Grading Rubric

Arithmetic expressions – 2 points

Type conversion and built-in functions – 2 points

math module functions – 2 points

random module usage – 2 points

print() formatting – 1 point

Code clarity and organization – 1 point

Total: 10 points